Photoinduced Electron Transfer in Hydrogen Bonded Donor-Acceptor Systems under Vibrational Strong Coupling

Summer Internship Report

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DECLARATION

I hereby declare that I have completed an internship under the guidance of Dr. Anoop Thomas from the Quantum Light Matter Science (QLiMS) Group, Indian Institute of Science (IISc), Bangalore. I am submitting a brief overview of my project to the National Institute of Science Education and Research (NISER), Bhubaneswar.

Signature of the Student

Date:

The internship work reported in this document entitled **Photoinduced Electron Transfer in Hydrogen Bonded Donor-Acceptor Systems under Vibrational Strong Coupling** was carried out under my supervision.

Signature of the Supervisor

Date:

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ABSTRACT

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1 Introduction

Photoinduced electron transfer (PET) is a fundamental process in photochemistry, playing a crucial role in natural and artificial systems such as photosynthesis, solar energy conversion, and molecular electronics. In many systems, hydrogen-bonded networks play a crucial role in facilitating efficient electron transfer between donor and acceptor molecules. The dynamics of such charge-transfer processes is typically governed by the interplay of electronic coupling, solvent interactions, and molecular vibrations.

The kinetic analysis of photoinduced electron transfer (PET) reactions in solution has been profoundly shaped by the Rehm-Weller model [21], which emerged from extensive studies of fluorescence quenching phenomena. This empirical framework describes how bimolecular electron transfer rate constants (k_q) vary with the thermodynamic driving force, showing two characteristic regimes: in the normal region, the rate increases with exergonicity, while beyond a certain driving force threshold, the rate plateaus at the diffusion limit [26]. This saturation behavior, first systematically quantified by Rehm and Weller in their 1970 landmark study, stood in apparent contradiction to theoretical expectations, as it failed to show the rate decrease predicted for highly exergonic reactions. The success of the model in interpreting the experimental data established diffusion as a dominant factor that masks the intrinsic kinetics of electron transfer in flexible donor-acceptor systems[1].

The resolution of this paradox came from Marcus theory [14, 15], which provides the fundamental physical basis for electron transfer processes. Developed in the 1950s-60s and later extended to photoinduced reactions[13], Marcus theory describes electron transfer rates through three parameters: the driving force (ΔG°), the reorganization energy (λ) and the electronic coupling (H_{el}). The most striking prediction of the theory, the inverted region where rates decrease with increasing exergonicity beyond $\Delta G^\circ = -\lambda$, was initially met with skepticism due to its absence in most of the solution phase studies involving PET[3]. This discon-

nect between theory and experiment persisted until the researchers recognized that the Rehm-Weller behavior represents a diffusion-limited special case of the Marcus theory[23], where the averaging over donor-acceptor distances in bimolecular encounters obscures the inverted region.

Hydrogen-bonded donor-acceptor complexes resolved this long-standing dichotomy by allowing simultaneous observation of both kinetic regimes[19]. In these systems, a fraction of molecules exist as associated complexes with fixed separation distances, while the remainder diffuse freely in solution. As demonstrated in the foundational work by Gopidas and Prasad[19], the associated fraction exhibits classic Marcus behavior including the inverted region, with electron transfer rates well described by the theoretical expression containing the exponential dependence on $(\Delta G^\circ + \lambda)^2$ [15]. Meanwhile, the diffusing fraction shows characteristic Rehm-Weller kinetics[21], with rates reaching but not exceeding the diffusion limit. This dual behavior within a single experimental system confirmed that the inverted region is indeed intrinsic to electron transfer physics, but becomes observable only when distance fluctuations are minimized[17].

Recent advances in quantum electrodynamics have demonstrated that vibrational strong coupling (VSC) can modulate chemical reactivity by hybridizing molecular vibrations with confined electromagnetic fields in optical cavities. This creates polaritonic states that alter potential energy surfaces, potentially impacting electron transfer dynamics. Understanding how VSC influences PET, particularly in hydrogen-bonded systems, could provide new insights into controlling charge-transfer processes without direct chemical modification.

This project investigates how VSC affects PET in hydrogen-bonded complexes. By coupling specific vibrational modes to a cavity field, we aim to explore whether electron transfer dynamics can be enhanced, suppressed, or otherwise modified through light-matter interactions. The study combines principles from photochemistry and quantum optics to uncover new mechanisms for controlling charge transfer processes

under strong coupling conditions.

2 Photoinduced Electron Transfer in Hydrogen-Bonded Complexes

Photoinduced electron transfer (PET) in hydrogen-bonded complexes represents an important class of reactions that bridge molecular photochemistry and biological electron transfer processes. The Marcus theory provides the fundamental framework for understanding the kinetics of PET, describing the electron transfer rate constant as a function of the driving force (ΔG°), reorganization energy (λ) and the electronic coupling (H_{el}) [13]. This theory predicts the characteristic inverted region where the rate decreases with increasing exergonicity, a characteristic rarely observed in bimolecular systems due to diffusion limitations [21]. Hydrogen-bonded donor-acceptor systems offer unique advantages for the study of PET, as they maintain fixed molecular orientations while allowing electronic coupling through the hydrogen bond network. The work of Prasad and Gopidas demonstrated that such systems can simultaneously exhibit both Marcus-type behavior in the associated complexes and Rehm-Weller kinetics for the diffusing species [19]. This dual behavior arises because hydrogen-bonded complexes undergo unimolecular electron transfer at fixed distances, while unassociated molecules follow conventional bimolecular quenching pathways.

The directional nature of hydrogen bonds makes them particularly effective mediators of electron transfer, often serving as superior conduits compared to through-space interactions [2]. This property, combined with their prevalence in biological systems, explains the continued interest in hydrogen-bonded PET systems as models for natural processes and as potential components in molecular electronics [27].

The integration of hydrogen-bonded PET systems with quantum optical cavities represents a particularly promising direction, where vibrational strong coupling has been shown to modify both electronic coupling (H_{el}) and reorganization energy (λ) through the formation of hybrid light-matter states[6].

3 Fluorescence Quenching in Electron Transfer Reactions

The study of fluorescence quenching mechanisms provides fundamental insights into the dynamics of photoinduced electron transfer processes, particularly in hydrogen-bonded donor-acceptor systems. These quenching phenomena can be broadly categorized into two distinct but often concurrent pathways: dynamic quenching arising from diffusion-controlled encounters between excited donors and free acceptors, and static quenching occurring within preassociated hydrogen-bonded complexes. The kinetic distinction between these pathways was first systematically characterized by Rehm and Weller through their seminal studies of bimolecular electron transfer reactions, where they established that diffusion-limited processes typically mask the intrinsic inverted region behavior predicted by Marcus theory[21]. This framework was later expanded by Valeur and Berberan-Santos in their authoritative treatment of molecular fluorescence, which demonstrated how time-resolved measurements could disentangle these competing quenching mechanisms through analysis of biexponential decay profiles[25].

In hydrogen-bonded systems, the static quenching component exhibits particularly rich behavior due to the unique electronic coupling facilitated by the hydrogen bond interface. The work of Prasad and Gopidas provided definitive evidence for this effect, showing that carboxylic acid dimers with well-defined 9.4 Å donor-acceptor separations mediate electron transfer with coupling strengths (H_{el}) of 5-50 cm^{-1}, sufficient to observe Marcus type driving force dependence including the inverted region [19]. This stands in contrast to the dynamic quenching component observed for unassociated molecules in the same system, which follows the characteristic Rehm-Weller saturation behavior at high exergonicities. The simultaneous observation of both regimes within a single experimental system offers powerful validation of Marcus theory's fundamental predictions, while also highlighting the critical role of molecular organization in electron transfer processes.

The physical origin of quenching in hydrogen-bonded systems involves

several coupled dynamical processes that occur across distinct timescales. Femtosecond infrared spectroscopy studies by Zhong and coworkers have revealed that hydrogen bond reorganization precedes electron transfer, with characteristic time constants of approximately 80 fs for the initial hydrogen bond adjustment and 1-10 ps for the subsequent electron transfer event [28]. This temporal hierarchy reflects the dual role of hydrogen bonds as both structural scaffolds and active participants in the electron transfer process.

The The fluorescence lifetime profiles of hydrogen-bonded donor-acceptor systems provide critical insights into the competing electron transfer pathways. In these systems, the decay kinetics typically exhibit bi-exponential behavior when both associated and unassociated species coexist in solution, as first quantitatively described by Lakowicz in his foundational work on time-resolved fluorescence spectroscopy [11]. The short-lived component ($\tau_1 \approx 1$–10 ns) corresponds to electron transfer within the hydrogen-bonded complexes, where the fixed donor-acceptor distance enables efficient coupling and rapid charge separation. This component follows first-order kinetics described by:

$$I_1(t) = A_1 e^{-t/\tau_1}$$

where the amplitude A_1 is proportional to the fraction of associated complexes, typically 10–30 % at millimolar quencher concentrations based on association constants $K_a \approx 10^2$–10^3 M^{-1} [19]. The lifetime τ_1 directly yields the electron transfer rate constant through:

$$k_{\mathrm{et}} = \frac{1}{\tau_1} - \frac{1}{\tau_0}$$

where τ_0 represents the unquenched donor lifetime (typically 50–200 ns for aromatic probes like pyrene or anthracene derivatives). In contrast, the long-lived component ($\tau_2 \approx 20$–90% of τ_0) arises from unassociated donors that undergo bimolecular quenching with free acceptors. This decay follows the time-dependent Stern-Volmer equation:

$$I_2(t) = A_2 e^{-t/\tau_2} - 2k_q[Q]\sqrt{\frac{Dt}{\pi}}$$

where k_q is the diffusion-controlled bimolecular rate constant (typically 1–5 $\times$ 10^{10} $M^{-1}s^{-1}$ in low-viscosity solvents), [Q] the quencher concentration, and D, the mutual diffusion coefficient [25]. The relative amplitudes of these components (A_1/A_2) directly reflect the association equilibrium, varying systematically with temperature and solvent polarity.

3.1 Analysis of Fluorescence Decay Kinetics

The temporal profiles of fluorescence decay provide critical insights into the molecular environments and electron transfer pathways within hydrogen-bonded donor-acceptor systems. These decays manifest as either monoexponential or biexponential functions, each revealing distinct aspects of the photophysical dynamics.
A monoexponential decay, characterized by the function:

$$I(t) = I_0 e^{-t/\tau}$$

indicates the presence of a single, homogeneous population of fluorophores experiencing identical quenching conditions.
A biexponential decay, described by:

$$I(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2}$$

signifies the coexistence of two distinct fluorophore populations with differential quenching environments. (τ_1 signifies short-lived component due to donors within hydrogen bonded complexes and τ_2 signifies long-lived component due to Unassociated donors undergoing collisional quenching).

4 Quantum Nature of Light-Matter Interactions in Optical Cavities

The quantum mechanical description of light as a harmonic oscillator provides the fundamental framework for understanding cavity quantum electrodynamics[4]. The electromagnetic field in an optical cavity can be quantized as a quantum harmonic oscillator, where each cavity mode of frequency ω corresponds to discrete energy states $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, with n representing the photon number and the zero point energy, $\frac{1}{2}\hbar\omega$, giving rise to measurable vacuum fluctuations [8]. These vacuum fluctuations, arising from Heisenberg's uncertainty principle $\Delta x\,\Delta p \geq \frac{\hbar}{2}$, manifest as non-zero electric field amplitudes even in the absence of photons.

The confinement of electromagnetic fields in optical cavities imposes specific boundary conditions that lead to the formation of standing wave patterns. For a Fabry-Pérot cavity consisting of two parallel mirrors separated by distance L, the boundary conditions require the electric field to vanish at the mirror surfaces, resulting in discrete allowed modes with wavevectors $k_n = \frac{n\pi}{L}$ $(n = 1, 2, 3, \ldots)$. These boundary conditions produce standing wave solutions of the form:

$$E_n(z,t) = E_0(a_n + a_n^\dagger)\sin(k_n z)e^{-i\omega_n t}$$

where $\omega_n = \frac{n\pi c}{L}$ are the discrete mode frequencies and c the speed of light [16]. The frequency spacing between consecutive modes, known as the free spectral range (FSR), is given by $\Delta\omega = \frac{\pi c}{L}$, while the spectral width of each mode is characterized by its full width at half maximum (FWHM) $\delta\omega = \frac{\omega}{Q}$, where Q is the cavity quality factor [6].

When molecules are placed within such a cavity, their interaction with the confined electromagnetic field leads to light-matter coupling. This coupling becomes particularly strong when the molecular transition frequency ω_m matches a cavity mode frequency ω_c, satisfying the resonant condition $\omega_m \approx \omega_c$ [22]. The coupling strength g between a single molecule and the cavity mode depends on the molecular transition dipole moment μ and the vacuum field amplitude E_0 through

$g = \mu E_0/\hbar$ [5].

In the strong coupling regime, achieved when the collective coupling strength $\Omega = g\sqrt{N}$ (for N molecules) exceeds both the cavity decay rate κ and molecular linewidth γ (i.e., $\Omega > (\kappa+\gamma)/2$), the system forms hybrid light-matter states called polaritons [18]. These polaritons appear as two new eigenstates separated in energy by the Rabi splitting $\hbar\Omega$, observable in spectroscopic measurements as an avoided crossing between the cavity and molecular resonances [12]. For vibrational strong coupling (VSC), this typically involves molecular vibrations (e.g., C=O stretch at 1700 cm^{-1}) coupling to cavity modes with Rabi splittings of 100–500 cm^{-1} [20]. Infrared spectroscopy serves as the principal experimental technique for investigating vibrational strong coupling phenomena, providing direct access to the hybrid light-matter states formed when molecular vibrations interact with confined electromagnetic fields. Fourier-transform infrared (FTIR) spectroscopy reveals the characteristic anti crossing behavior of polariton formation, where the Rabi splitting manifests as two distinct absorption bands separated by $\hbar\Omega$, with the original molecular vibration and cavity mode resonances disappearing due to the formation of new eigenstates [7]. The spectral resolution of these measurements is determined by the full width at half maximum (FWHM) of the absorption features, which reflects both the intrinsic linewidth of the molecular vibration (typically 5-20 cm^{-1} for fundamental transitions) and the quality factor of the cavity mode ($\delta\nu \approx \nu/Q$) [16]. For precise determination of coupling strengths, the FWHM of both the cavity mode and molecular absorption must be carefully characterized, as the condition for strong coupling requires that the Rabi splitting Ω exceeds the average of these linewidths [6]. The fabrication of Fabry-Pérot cavities for vibrational strong coupling involves precise control over both the cavity geometry and the optical properties of the embedded medium. A typical experimental configuration consists of two highly reflective mirrors separated by a polymer matrix containing the molecules of interest, with the cavity length L determined by the polymer thickness [22]. The polymer matrix serves multiple critical functions: it provides mechanical support for maintain-

ing mirror parallelism, determines the cavity's effective refractive index n (typically 1.4–1.6 for common polymers like PMMA), and controls the molecular concentration N available for coupling [5]. The resonant condition for coupling to a specific molecular vibration at frequency ν_0 requires the cavity length to satisfy $L = m\lambda_0/2n$, where $\lambda_0 = c/\nu_0$ and m is the mode order. For first-order coupling ($m = 1$) to the C=O stretch at 1700 cm^{-1} ($\lambda_0 \approx 5.88$ μm) in a polymer with $n = 1.5$, this yields $L \approx 1.96$ μm.

The free spectral range (FSR) of such a cavity, given by $\Delta\nu = c/2nL$, is inversely proportional to both the refractive index and the physical thickness of the polymer layer [18]. For the example above ($n = 1.5$, $L = 1.96$ μm), the FSR calculates to approximately 5100 cm^{-1}, ensuring isolation of the target vibrational mode from adjacent cavity modes. The spin-coating process used to deposit the polymer matrix allows precise control over the cavity thickness through variations in spin speed and time—higher spin speeds (3000–6000 rpm) and longer spin times (30–60 s) produce thinner, more uniform films, thereby increasing the FSR [10].

A practical Fabry-Pérot cavity for vibrational strong coupling experiments consists of two parallel mirrors separated by the polymer matrix containing the target molecules [24]. The bottom mirror is typically a metallic layer (e.g., 100 nm gold) evaporated onto a substrate, while the top mirror can be either another metallic layer or a dielectric Bragg stack, chosen to have high reflectivity ($> 99\%$) in the relevant infrared spectral region [9]. The standing wave pattern formed between the mirrors creates a spatially varying coupling strength, with molecules located at the antinodes experiencing maximum interaction with the cavity field.

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